LAB EXPERIMENT RECORD

Set 30: Dataset: Mobile_App_Usage.csv

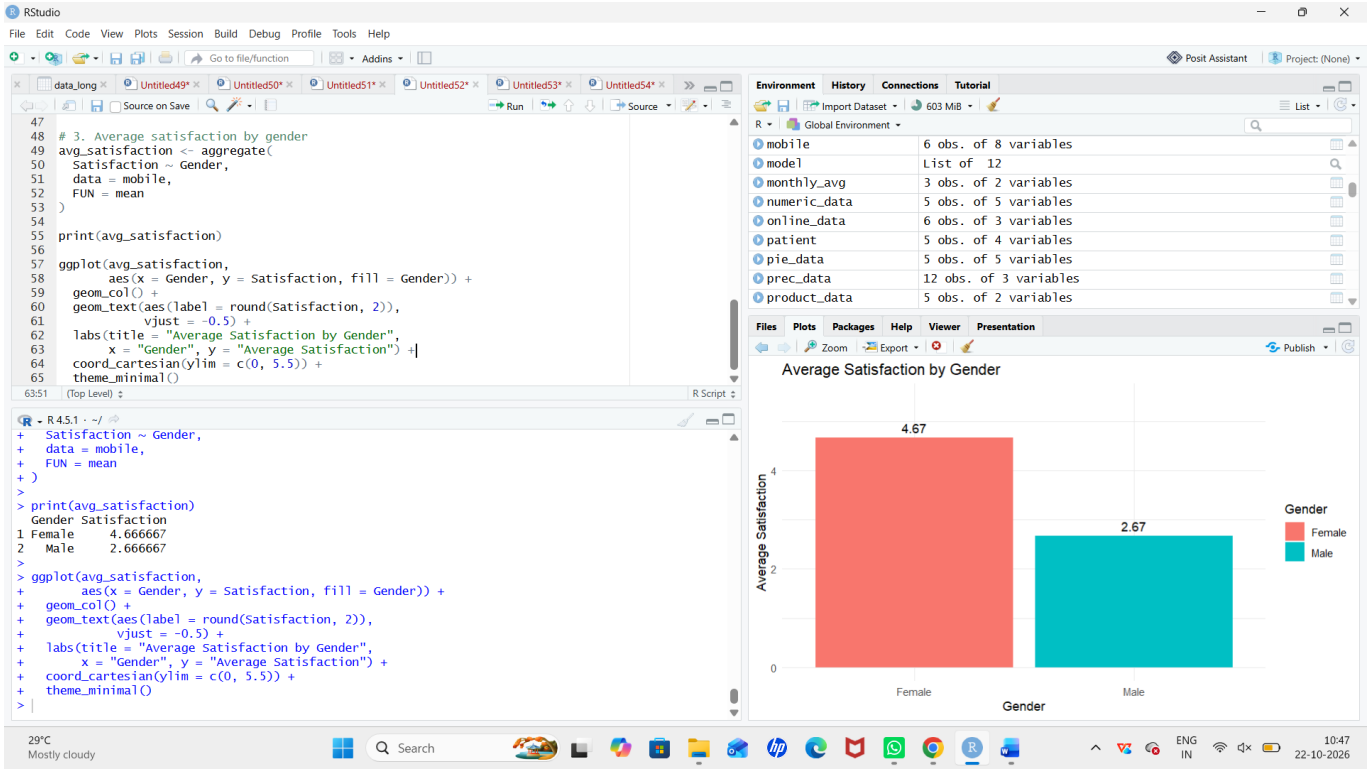
Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
Date	22/10/2026
Day	Thursday

AIM

To analyze mobile application usage using R and Tableau by creating a histogram, density plot, scatter plot, correlation analysis, gender-wise satisfaction bar chart, and interactive dashboard.

1.

```
# Create the dataset
mobile <- data.frame(
  User_ID = c("U01", "U02", "U03", "U04", "U05", "U06"),
  Gender = c("Male", "Female", "Male", "Female", "Male", "Female"),
  Age = c(20, 22, 19, 21, 23, 20),
  Screen_Time = c(4.5, 6.0, 3.2, 7.1, 2.8, 5.4),
  App_Usage_Count = c(18, 25, 12, 30, 10, 22),
  Data_Used = c(2.4, 3.8, 1.6, 4.5, 1.2, 3.1),
  Satisfaction = c(3, 5, 3, 5, 2, 4),
  Usage_Date = as.Date(c(
    "2025-01-08", "2025-01-08",
    "2025-02-11", "2025-02-11",
    "2025-03-14", "2025-03-14"
  ))
```



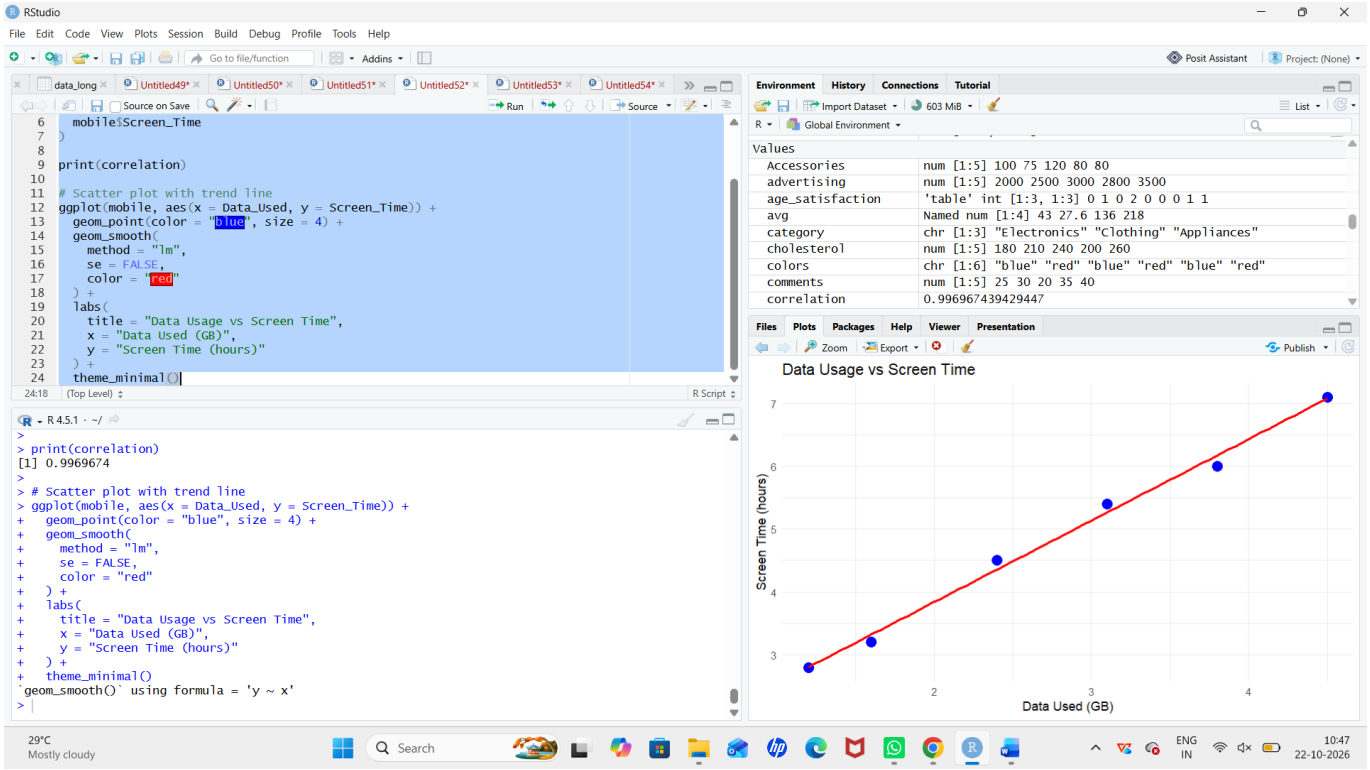
2.

```
library(ggplot2)

# Calculate correlation
correlation <- cor(
  mobile$Data_Used,
  mobile$Screen_Time
)

print(correlation)

# Scatter plot with trend line
ggplot(mobile, aes(x = Data_Used, y = Screen_Time)) +
  geom_point(color = "blue", size = 4) +
  geom_smooth(
    method = "lm",
    se = FALSE,
    color = "red"
  ) +
  labs(
    title = "Data Usage vs Screen Time",
    x = "Data Used (GB)",
```



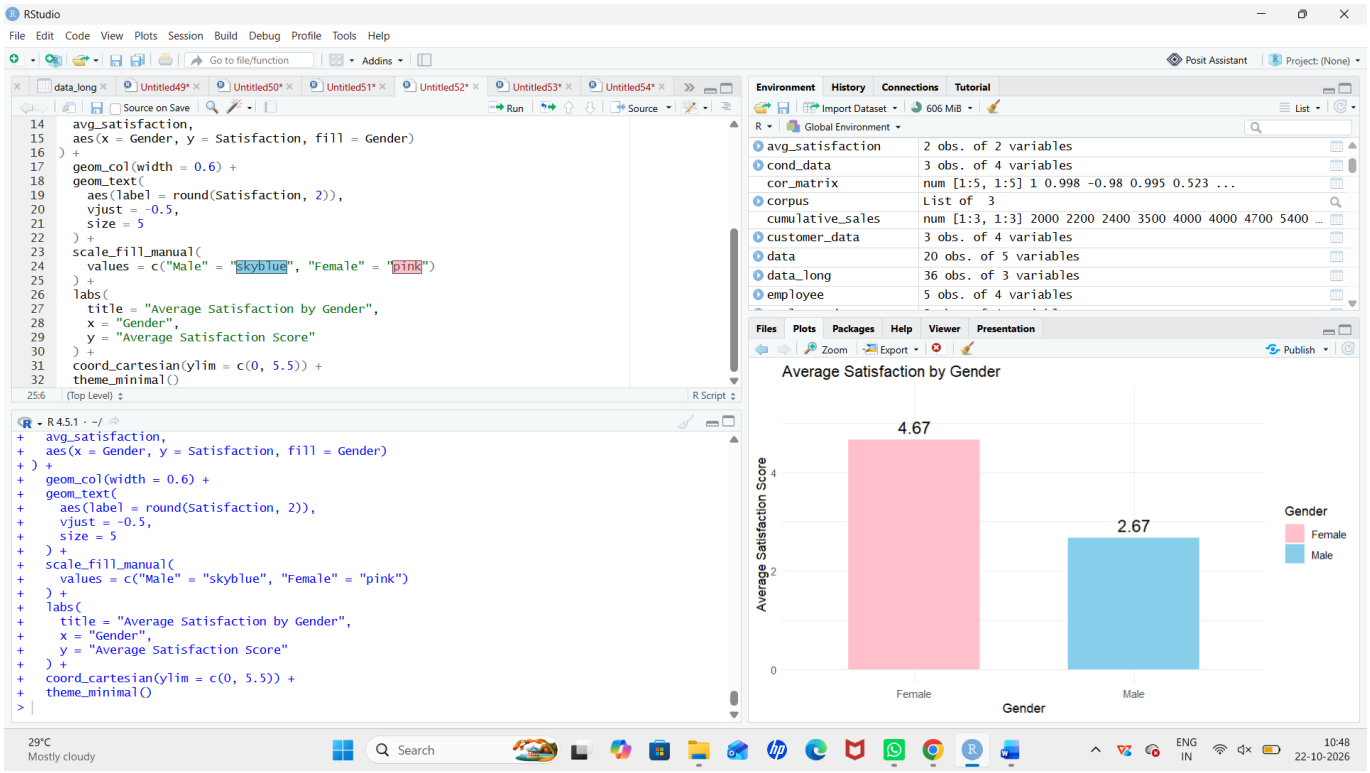
3.

```
library(ggplot2)

# Calculate average satisfaction
avg_satisfaction <- aggregate(
  Satisfaction ~ Gender,
  data = mobile,
  FUN = mean
)

print(avg_satisfaction)

# Bar chart with value labels
ggplot(
  avg_satisfaction,
  aes(x = Gender, y = Satisfaction, fill = Gender)
) +
  geom_col(width = 0.6) +
  geom_text(
    aes(label = round(Satisfaction, 2)),
    vjust = -0.5,
    size = 5
  ) +
  scale_fill_manual(
    values = c("Male" = "skyblue", "Female" = "pink")
  ) +
  labs(
    title = "Average Satisfaction by Gender",
    x = "Gender",
    y = "Average Satisfaction Score"
  ) +
  coord_cartesian(ylim = c(0, 5.5)) +
  theme_minimal()
```


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
avg_satisfaction,
aes(x = Gender, y = Satisfaction, fill = Gender)
) +
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title = "Average Satisfaction by Gender",
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theme_minimal()
Environment History Connections Tutorial
avg_satisfaction 2 obs. of 2 variables
cond_data 3 obs. of 4 variables
cor_matrix num [1:5, 1:5] 1 0.998 -0.98 0.995 0.523 ...
corpus List of 3
cumulative_sales num [1:3, 1:3] 2000 2200 2400 3500 4000 4000 4700 5400 ...
customer_data 3 obs. of 4 variables
data 20 obs. of 5 variables
data_long 36 obs. of 3 variables
employee 5 obs. of 4 variables
Files Plots Packages Help Viewer Presentation
Average Satisfaction by Gender
4.67
2.67
Average Satisfaction Score
Female
Male
Gender
Gender
Female
Male
29°C
Mostly cloudy
Search
ENG
IN
10:48
22-10-2026

3. TABLEAU: INTERACTIVE SCATTERPLOT

A. Import the dataset

1. Save the data as Mobile_App_Usage.csv.
2. Open Tableau Desktop or Tableau Public.
3. Select Connect → Text File and choose the CSV file.
4. Confirm that Usage_Date is recognized as a Date field.

B. Create the bubble chart

1. Open a new worksheet and rename it Screen Time vs Data Used.
2. Drag Data_Used to Columns.
3. Drag Screen_Time to Rows.
4. Select Circle in the Marks dropdown.
5. Drag Gender to Color.
6. Drag App_Usage_Count to Size.
7. Drag User_ID to Detail.
8. Drag User_ID to Tooltip to identify each user.

The bubbles show the relationship between data usage and screen time. Larger bubbles represent more app usage.

C. Create the usage trend line chart

1. Open another worksheet and rename it Usage Trend.
2. Drag Usage_Date to Columns.
3. Drag Screen_Time to Rows.
4. Set the Marks dropdown to Line.
5. Optionally drag Gender to Color to compare usage trends by gender.

Because the dataset contains three dates, the chart shows only a short trend across January, February, and March 2025.

D. Add filters

1. Drag Gender to the Filters shelf and select Show Filter.
2. Drag Usage_Date to the Filters shelf and select Show Filter.
3. Choose a suitable filter display, such as a dropdown for Gender and a date range for Usage_Date.

E. Design the mini dashboard

1. Click New Dashboard.
2. Drag the Screen Time vs Data Used bubble chart onto the dashboard.
3. Drag the Usage Trend line chart onto the dashboard.
4. Show the Gender and Usage_Date filters.
5. Add the title Mobile App Usage Analysis Dashboard.
6. Test the filters to see how the displayed charts change.

Dashboard interpretation:

- U04 has the highest screen time (7.1 hours), data usage (4.5 GB), and app usage count (30).
- U05 has the lowest screen time (2.8 hours), data usage (1.2 GB), and app usage count (10).
- Female users have a higher average satisfaction score in this dataset.
- The bubble chart compares screen time and data usage, while the line chart compares screen time across the three usage dates.

PROCEDURE

1. Import or create the mobile app usage dataset in R.
2. Generate a histogram and density plot to examine screen-time distribution.
3. Create a scatter plot and calculate the correlation between data usage and screen time.
4. Calculate average satisfaction by gender and display it in a bar chart.
5. Import the CSV into Tableau and create a bubble chart and usage trend line chart.
6. Add Gender and Usage_Date filters and combine both charts into a mini dashboard.

RESULT

Mobile app usage was analyzed using R and Tableau. Screen time ranged from 2.8 to 7.1 hours, and data usage showed a strong positive correlation with screen time (approximately 0.98). Female users had a higher average satisfaction score than male users in the supplied sample. The Tableau mini dashboard enabled comparisons of screen time, data usage, app usage count, gender, and usage dates. These results describe the six recorded users and should not be generalized to all mobile app users.